

A Lightweight IoT Device-friendly Anomaly Sitting Posture Detector for Protecting Adolescent Bone Development

Haowen Ji, Jiacheng Xie

Duke Kunshan University/Duke University,
Emails: hj118@duke.edu, jx105@duke.edu

Peng Sun

Duke Kunshan University,
Email: peng.sun568@duke.edu

Abstract—In recent years, people have tended to spend a lot of time sitting while unconsciously having bad posture. Research has shown that bad sitting poses, for instance, sitting cross-legged, could lead to serious spinal disorders, including thoracic spine tilting and pelvic posterior tilting. Correspondingly, correcting bad sitting posture is essential to prevent people, especially adolescents, from those problems. Inspired by the recent prevalence of Internet-of-Things (IoT) health monitoring systems, we proposed a lightweight vision-based bad sitting pose detector, which can be applied to IoT devices with a camera on. This method consists of two main parts: 1) generating human pose landmarks of the subjects and 2) integrating pose landmarks with shoe detection data into a pose classifier. To verify the reliability of our method, we also created a dataset containing a total of 5042 labeled sitting postures. The proposed method achieves high accuracy in detecting bad sitting postures. Besides, adding the shoe's position as a supplementary attribute was found to be effective in detecting lower body bad poses. This paper demonstrates the feasibility of applying a pose detection model on IoT devices to effectively correct bad sitting postures in daily lives.

Index Terms—Anomaly pose detection, object detection, computer vision, health monitor.

I. INTRODUCTION

In recent years, the incidence of cervical spondylosis, lumbar spondylosis, and other spinal disorders has shown a younger trend [1]. This is highly related to the condition that students and employees are forced to spend more time studying and working in a sitting posture [2]. Among the common bad sitting postures, researchers have shown that sitting cross-legged for long hours has a negative impact on our pelvis and pulmonary function [3]. Consequently, it is increasingly crucial to take action to correct bad sitting postures and sedentary. However, it is difficult for people to remind themselves when they are experiencing harmful sitting postures and sedentary, especially when they are obsessed with their work. Inspired by the widely applied Internet-of-Things (IoT) health monitoring systems, we proposed a lightweight vision-based pose detection solution for reminding users when they perform bad sitting postures.

The vision-based algorithms that perform the task of pose estimation, such as the MoveNet Thunder [4], are usually 2D-based solutions. This means that for detecting human poses that involve overlapping body parts, such as crossing legs,

these vision-based algorithms would not generate reliable results since no depth information is provided. Therefore, in our task of detecting harmful sitting poses, which includes the head leaning forward and crossing legs when sitting, algorithms like MoveNet Thunder would not be able to provide correct pose estimation for crossing legs.

To address the above issues, we propose a lightweight anomaly posture detection method for IoT devices with limited computing power. Briefly, to overcome the lack of capability of existing posture detection methods to detect anomaly sitting postures in 3D conditions, our method significantly improves the detection accuracy of three common anomaly sitting postures, i.e., head-forward, cross-leg, and normal sitting, by identifying the target person's belongings. Additionally, a traditional pose classifier that is trained with human body parts landmarks of 17 body parts generated by the Movenet Thunder model is implemented on an edge device to explore the accessibility of utilizing the trained model in the real-world scenario. The main contributions of our research are summarized as the following:

- We proposed a lightweight anomaly posture detection method for accurately detecting cross-leg sitting without depth information.
- We proposed a novel method of integrating object detection results as supplemental information to the missing topological relationship between crossing human body parts.
- We also created a dataset containing a total of 5042 labeled sitting postures for supporting the further research.

The remainder of the this paper is organized as follows. In Section II, we first provide a brief survey regarding the existing pose detection methods. Then, in Section III, we will discuss details of the problem we consider in our design and introduce our method in detail. The experimental results will demonstrate in Section IV. Finally, Section V will conclude this paper.

II. RELATED WORK

A. IoT Application

The application of human pose estimation on IoT devices has been studied recently. Hong et al. [5] proposed a wearable

AI-based IoT system that identifies human postures. However, wearable devices have an “intrusive” feature [6]. The authors in [6] demonstrated that using computer vision techniques on human motion detection can provide a non-intrusive way of monitoring the safety of elderly individuals. Thus, applying computer vision-based human pose estimation on IoT devices would provide a non-intrusive way of detecting human motions.

B. Pose Estimation Tech

This would involve using pose estimation models, such as OpenPose [7], HRNet [8], and MoveNet [4]. OpenPose is widely used in the area of multi-person pose estimation tasks since its born in 2017. HRNet (High-Resolution Network) is a type of neural network architecture that aims to address the limitations of traditional neural networks that struggle with high-resolution inputs, which has shown impressive results in human pose estimation tasks. MoveNet designed to be lightweight and efficient, making it an ideal choice for human pose estimation tasks on IoT devices. Since IoT devices often have limited computational resources and power, using a model like MoveNet can help reduce the strain on these devices while still providing accurate pose estimation results. Additionally, the single-stage regression-based approach used by MoveNet helps to further reduce the computational cost of the model, making it well-suited for deployment on edge devices where processing power may be even more limited.

The design of using a single-stage regression-based approach to predict key points on the body determines MoveNet’s limitation in detecting pose estimation on overlapping body parts. SPM-Net [10] (Single-Stage Multi-Person Pose Machines), a model also uses the single-stage strategy as the MoveNet, also struggling with the overlapping body parts detection as Wei Yang and colleagues stated. Researchers put a lot of effort into addressing the overlapping body parts detection issue. The HigherHRNet [11] was proposed which achieves state-of-the-art performance on bottom-up multi-person pose estimation. Using enhanced channel-wise and spatial information, Jin et al. [12] proposed a new method for multi-person pose estimation that addresses the problem of overlapping body parts.

However, there is still a need for further research to address the issue of overlapping body parts in the context of human pose estimation on IoT devices, which would improve the accuracy and usability of these systems for non-intrusive monitoring of human motion. A research [9] shows that MoveNet is one of the fastest pose estimation models compared with OpenPose [7] and PoseNet [13]. In this study, we aim to address this research gap by proposing a new method for human pose estimation that is specifically designed to address the issue of overlapping body parts in the context of light weighted IoT devices.

III. THE PROPOSED METHOD

As mentioned above, the aim of this research is to develop a lightweight vision-based sitting pose detection method to

identify abnormal sitting postures, such as crossed leg posture and forward head postures. The proposed method will rely on computer vision techniques to analyze images or video feeds of seated individuals and classify their sitting postures as either normal or abnormal based on predefined criteria. The future goal of this research is to contribute to the development of IoT tools and technologies that can help individuals correct their posture and prevent potential health issues associated with prolonged sitting in abnormal postures.

The overall structure of our anomaly posture detection model is shown in Fig. 1. The method takes images as input and outputs the types of sitting postures. These images are first preprocessed into skeleton landmarks using the pose estimation model, MoveNet [4]. Meanwhile, these images are also feed into a shoe detector to generate the bounding box of shoes, which indicate the position of the shoes. A pose classifier is built based on the combination of skeleton landmarks and shoe position.

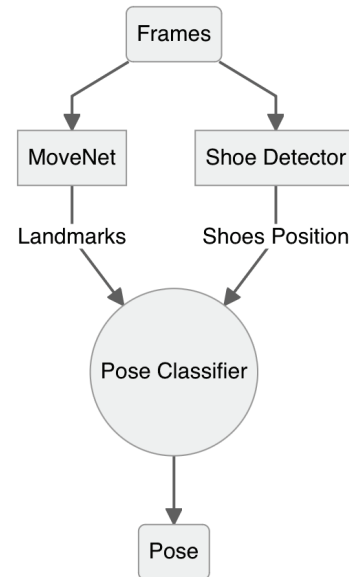


Fig. 1. The flow chart of the proposed method

A. MoveNet

MoveNet is a fast and accurate pose estimation model used to detect the keypoints of a human body. There are two versions of MoveNet, which are Lightning and Thunder. The accuracy-oriented model MoveNet Thunder is adopted in our design.

MoveNet Thunder takes images as input and outputs the position and score of 17 human skeleton nodes. The position is represented in a 2 dimensional by x and y, so a node is described by 3 features. Thus, MoveNet Thunder will output 51 features of the detected human skeleton. In this research, MoveNet Thunder is used to preprocess the input images and extract a series of human body keypoints from the images.

B. Shoe Detector

The TensorFlow Object Detection API is adopted for designing the shoe detector. The TensorFlow Object Detection API provides a set of pre-trained models that can be used for the object detection task. The EfficientDet-Lite0 [14] is chosen as the pre-trained model for the detection task in this method.

EfficientDet-Lite0 is a lightweight object detection model, which is designed to be fast and efficient, making it well-suited for IoT devices where computing and memory usage are important considerations. EfficientDet-Lite0 achieves high accuracy while using a small number of parameters, which makes it an ideal choice for deployment on edge devices with limited computational resources. Therefore, in our work, EfficientDet-Lite0 was applied to detect the shoes in the given images. This model outputs the bounding boxes of the shoes in the images, which show the position of the shoes.

C. Pose Classification Model

The pose classification model is a type of machine learning model that takes the landmark coordinates of a person in an image and predicts the pose category of that person. Pose landmarks are specific points on a person's body that can be extracted by the pose estimation method.

The input images are processed into 51 features of the detected human skeleton by the MoveNet Thunder pose estimation model. The shoe detector generates the bounding boxes of each input image, and the corresponding bounding boxes are represented by their center coordinates (xmid, ymid). The adding of center coordinates and detection scores as new features to the original dataset results in a total of 57 features. The resized dataset is used as an input for the pose classification model to classify the categories of anomaly sitting poses.

In our design, the pose classification model consists of two parts. The first part of the model calculates a pose embedding from the detected landmark coordinates. The second part of the model feeds pose embedding through several Dense layers to predict the pose class. The structure of the pose classification model used in this research is shown in Fig. 2, and detailed information about the layers are presented in Table I. The input layer has the same size as the landmark features, which is 57. Then in the Embedding layer, the pose center of the input data is changed to the origin, and the input landmark coordinates are scaled to the pose size as one and flattened into feature vectors with size 19×3 for pose classification. Two dense layers are then used in the model, each followed by one dropout layer. The two dense layers contain 128 and 64 neurons, and the dropout layers randomly drop out some of the activations with a 0.5 dropout rate. The last dense layer produces the probability distribution over the possible output classes, which are normal, crossed leg, and head forward. The activation function used in the first two dense layers is ReLU, and the final dense layer uses Softmax. To train the model, the data were randomly split into training (70%), validation (15%), and test (15%) sets.

The pose classification model is trained using a total of 200 epochs and a batch size of 16.

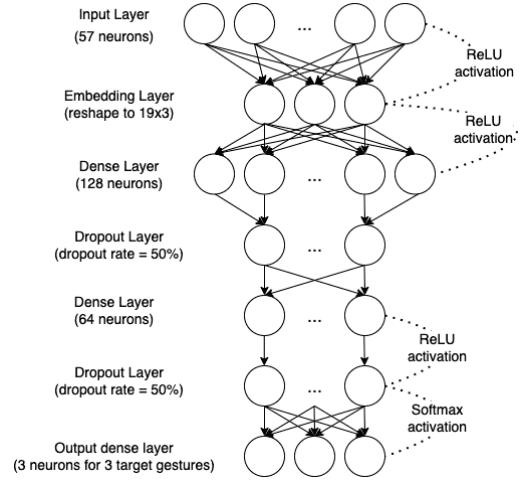


Fig. 2. The proposed pose classifier model

TABLE I
THE STRUCTURE OF THE POSE CLASSIFICATION MODEL

Layers	Parameters	Value
Input layer	Input size	57×1
Embedding layer	Output size	19×3
	Transfer function	ReLU
Dense layer	# of neurons	128
	Transfer function	ReLU
Dropout layer	Output size	128×1
	Dropout rate	0.5
Dense layer	# of neurons	64
	Transfer function	ReLU
Dropout layer	Output size	64×1
	Dropout rate	0.5
Dense layer	# of neurons	3
	Transfer function	Softmax

IV. PERFORMANCE EVALUATION

A. Experiment Environment

The real-world application of the abnormal pose estimation model is performed with the following hardware:

- Processor: Intel Core i7 (2.6 GHz, 6 cores)
- Graphic card: Intel UHD Graphics 630
- Memory: 16 GB

The classification model was trained on a virtual machine with the following spec:

- Processor: Intel(R) Xeon(R) CPU E5-2660 v3 @2.60GHz (16 cores)
- Graphic card: TITAN RTX
- Memory: 64 GB
- OS: Ubuntu 20.04.3 LTS
- CUDA Version: 11.2

We used Jupyter Notebook to run Python scripts, and Table II shows our Python version and versions of the main Python packages.

TABLE II
PYTHON ENVIRONMENT CONFIGURATION

Python	3.8.13
Matplotlib	3.5.1
Numpy	1.22.3
Opencv-python	4.6.0.66
Tensorflow	2.10.0

B. Dataset

For evaluating our proposed method, we created a dataset comprising of 5,042 labeled images. The images were captured from multiple videos recorded by an iPad mini, with each video ranging from one to two minutes in length. Participants in the videos were requested to adopt three distinct postures - normal posture, crossed leg posture, and forward head posture - which were subsequently extracted frame by frame to form the dataset. Each frame of the video depicts a full-bodied person in a particular posture, resulting in 2511 images of the cross-leg posture, 1247 images of the forward head posture, and 1117 images of the normal posture. The examples of these three posture categories are displayed in Fig. 3. The images were taken from an overhead perspective and were shot from the side. Other scenes containing similar images were also included in the dataset, which were subsequently categorized into their respective posture categories and placed in separate folders.

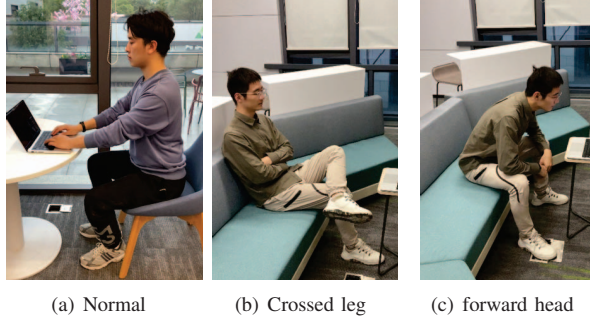


Fig. 3. Examples of the original dataset. (a) An example of the normal sitting posture; (b) An example of the crossed leg posture; and (c) The forward head posture

C. Performance Metrics

For evaluating the performance of classification models, a confusion matrix is a commonly adopted evaluation metric. The matrix provides a comprehensive breakdown of the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) generated by the model. In the current study, the confusion matrix was employed to evaluate the outcomes of the abnormal sitting pose detection method. Table III illustrates an example of the confusion matrix utilized in this evaluation process.

The metrics for performances derived from the confusion matrix are as follows:

TABLE III
PYTHON ENVIRONMENT CONFIGURATION

	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)

- For accuracy, it shows the proportion of correctly classified observations, which is calculated as:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}. \quad (1)$$

- For precision, it shows the proportion of true positives in the total of positive predictions, which is calculated as:

$$Precision = \frac{TP}{TP + FP}. \quad (2)$$

- For recall, it shows the proportion of positive cases correctly predicted, which is calculated as:

$$Recall = \frac{TP}{TP + FN}. \quad (3)$$

- For f1 - score, it combines precision and recall, which is calculated as:

$$F1score = \frac{2(P \times R)}{P + R}. \quad (4)$$

D. Sitting Pose Detection Performance

Table IV displays the outcome of the abnormal sitting pose detection method, indicating an accuracy of 97.27%. An examination of the matrix and Table IV reveals an enhancement in precision for the cross-leg category, as well as an increase in recall for the normal posture category by 0.5. The analysis of the confusion matrix further suggests a decline in misclassification of normal postures as the crossed leg posture, leading to the inference that improved shoe detection has contributed to the better detection of lower body postures, thereby elevating overall detection accuracy.

TABLE IV
CONFUSION MATRIX RESULT OF THE SITTING POSE DETECTION

	Precision	Recall	F1 score
Crossed leg	0.93	1.00	0.97
Forward head	1.00	1.00	1.00
normal	1.00	0.91	0.95

E. Ablation studies

In order to evaluate the efficacy of augmenting the model with object detection as an attribute, we conducted an experiment to develop a baseline sitting pose detection model without the shoe detector. The same training and evaluation procedures were applied to this model, and the outcomes are presented in Table IV. The baseline model achieved an accuracy of 95.63%, with relatively low precision for the cross-leg category and low recall for the normal posture category, indicating some misclassification of normal postures

as crossed-leg postures. This misclassification was likely due to incorrect landmarks generated by the pose estimation preprocessor. Nonetheless, the forward head posture category exhibited good results, suggesting that the baseline model already attained a relatively high level of performance.

TABLE V
CONFUSION MATRIX RESULT OF THE SITTING POSE DETECTION

	<i>Precision</i>	<i>Recall</i>	<i>F1 score</i>
Crossed leg	0.90	1.00	0.95
Forward head	1.00	0.99	0.99
normal	1.00	0.86	0.93

To illustrate the improved performance of our method relative to the baseline model, we present an example in Fig. 4, which showcases the incremental gains of incorporating a shoe detector. This example include the case in which parts of the feet were occluded, leading to inadequate detection by MoveNet, as depicted in the skeleton. Our comparison indicates that 36 images were accurately classified using the new classifier, thereby demonstrating the efficacy of incorporating the shoe detector attribute.

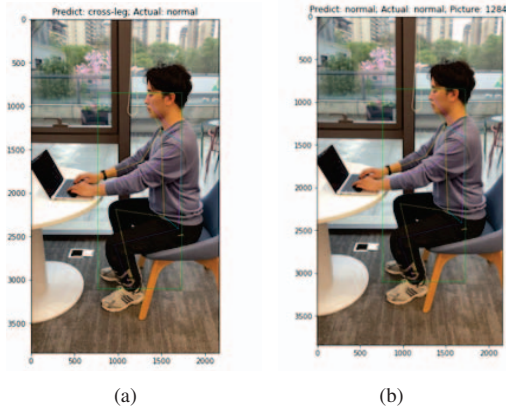


Fig. 4. Examples of the classification result of the baseline method and our method. The (a) shows the wrong sitting pose detection by the baseline model. The (b) shows the correct detection on the same image by our model

F. Model Application

The performance of the model application program was evaluated through real-time pose classification. For this evaluation, we used a pre-recorded testing video captured with the same camera device as used in the data collection stage, ensuring compatibility with the input format required by the MoveNet model and avoiding any discrepancies in frame size that could impact performance. During the performance evaluation stage, the model was applied in real-time using the MacBook's built-in camera, which has a frame size of 720×1280. The evaluation environment was designed to reflect real-world scenarios, and the tests were conducted in dormitories, characterized by cluttered backgrounds and uneven lighting distribution. Moreover, the camera was posi-

tioned a little farther away from the subject to challenge pose detection.

The evaluation results were generally positive. One example is shown in Fig. 5, 6, 7. The classifier accurately detected the subject's pose, and the pose labels were correctly assigned. These findings suggest that the sitting pose detection method can be deployed on edge devices and can perform well in complex environments.

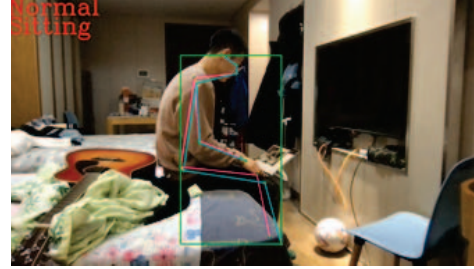


Fig. 5. Example of the successful classification of normal pose

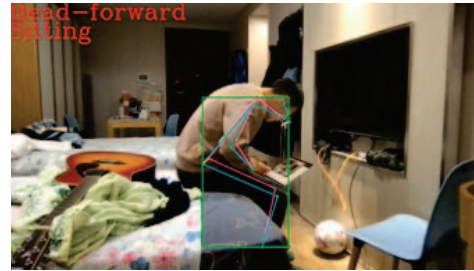


Fig. 6. Example of the successful classification of forward head pose

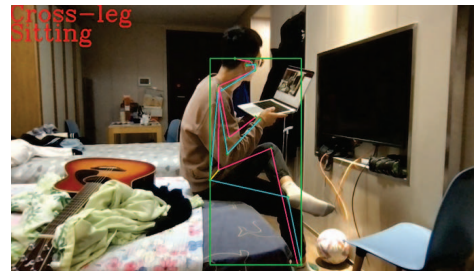


Fig. 7. Example of the successful classification of crossed leg pose

As demonstrated in Fig. 7, the human pose prediction generated by the pose estimation model cannot indicate the legs are crossed, indicating the lack of topological relationship between the subject's left and right legs. We posit that this issue results from the absence of depth information, as the MoveNet Thunder model utilized single-stage strategy to detect human body parts. Noticeably, this inaccuracy is overcome by our proposed method of adding shoes positions to give supplementary information.

V. CONCLUSION AND FUTURE WORK

This research proposes a novel lightweight detection model to identify abnormal sitting postures, and evaluates its performance in real-time human pose analysis on desktop platforms, enabling its use in real-world IoT scenarios. To address limitations in existing pose estimation models, we augment the detection model with a shoe detector that provides supplementary information on foot position and enhances the localization of legs. Compared to the baseline classifier, our model achieves a 1.64% increase in accuracy and successfully rectifies half of the misclassified labels. Empirical evaluation on video data demonstrates the effectiveness of our model in daily-life scenarios.

The vision-based lightweight abnormal sitting pose detection method has multifaceted implications. Primarily, it provides a novel approach to addressing posture-related health concerns. By enabling real-time posture monitoring, the method may assist in mitigating musculoskeletal disorders, spinal deformities, and other associated health conditions. Moreover, due to the lightweight nature of the method, it is highly suitable for individuals to monitor their posture, enabling timely feedback on sitting habits and posture. This allows individuals to modify their posture and sitting habits, leading to long-term improvements in health status. This technology is amenable to integration into IoT-based applications such as smart chairs and wearable devices, facilitating a seamless monitoring experience for users.

In terms of future work, the shoe detection component of the proposed model could be improved by utilizing a larger and more diverse dataset of shoes to enhance the performance of the object detector. Currently, the restricted selection of shoe types used in the training process limits the generalizability of the model. Additionally, the model could benefit from exploring the use of alternative approaches that solely consider the lower body landmarks to avoid the potential interference from the upper body landmarks. Specifically, two models were developed in this study that solely utilized the pose of the lower body. The first model used eighteen dimensions that exclusively consisted of the lower body landmarks. However, its test accuracy was low, and most images were classified as cross-postures, potentially due to insufficient features included in the classifier. The second model utilized fifty-one features, including both upper and lower body landmarks, and resulted in accuracy comparable to the baseline model. Therefore, further exploration of models that solely utilize the lower body keypoints may be an area for future research.

REFERENCES

- [1] S. Reineck, R. G. Hazard, K. Coleman, and M. Pope, "A continuous passive lumbar motion device to relieve back pain in prolonged sitting," *Advances in industrial ergonomics and safety IV*, pp. 971–976, 2002.
- [2] H. L. Xie, "Analysis of prevalence rate and related factors of adolescent myopia," *Chinese Medical Journal*, 2010.
- [3] J. T. Sun, M. An, X. B. Yan, G. H. Li, and D. B. Wang, "Prevalence and Related Factors for Myopia in School-Aged Children in Qingdao," *Journal of Ophthalmology*, vol. 2018, pp. 1–6, 2018, doi: <https://doi.org/10.1155/2018/9781987>.
- [4] R. Votel and N. Li, "Next-Generation Pose Detection with MoveNet and TensorFlow.js," 2021. <https://blog.tensorflow.org/2021/05/next-generation-pose-detection-with-movenet-and-tensorflowjs.html>
- [5] Z. Hong, M. Hong, N. Wang, Y. Ma, X. Zhou, and W. Wang, "A wearable-based posture recognition system with AI-assisted approach for healthcare IoT," *Future Generation Computer Systems*, Sep. 2021, doi: <https://doi.org/10.1016/j.future.2021.08.030>.
- [6] Z. Luo et al., "Computer vision-based descriptive analytics of seniors' daily activities for long-term health monitoring," *Machine Learning for Healthcare (MLHC)*, vol. 2, no. 1, 2018.
- [7] Z. Cao, G. Hidalgo Martinez, T. Simon, S.-E. Wei, and Y. A. Sheikh, "OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 7291–7299, 2019, doi: <https://doi.org/10.1109/tpami.2019.2929257>.
- [8] J. Wang et al., "Deep High-Resolution Representation Learning for Visual Recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 10, pp. 3349–3364, 2020, doi: <https://doi.org/10.1109/tpami.2020.2983686>.
- [9] B. Jo and S. Kim, "Comparative Analysis of OpenPose, PoseNet, and MoveNet Models for Pose Estimation in Mobile Devices," *Traitement du Signal*, vol. 39, no. 1, pp. 119–124, Feb. 2022, doi: <https://doi.org/10.18280/ts.390111>.
- [10] X. Nie, J. Feng, J. Zhang, and S. Yan, "Single-Stage Multi-Person Pose Machines," 2019 *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 6950–6959, 2019, doi: <https://doi.org/10.1109/ICCV.2019.00705>.
- [11] B. Cheng, B. Xiao, J. Wang, H. Shi, T. S. Huang, and L. Zhang, "HigherHRNet: Scale-Aware Representation Learning for Bottom-Up Human Pose Estimation," 2020 *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5385–5394, 2020, doi: <https://doi.org/10.1109/CVPR42600.2020.00543>.
- [12] K. Su, D. Yu, Z. Xu, X. Geng, and C. Wang, "Multi-Person Pose Estimation With Enhanced Channel-Wise and Spatial Information," 2019.
- [13] D. Oved, "Real-time Human Pose Estimation in the Browser with TensorFlow.js," *TensorFlow.js Blog*, 10 May 2018. <https://blog.tensorflow.org/2018/05/real-time-human-pose-estimation-in.html>.
- [14] "TensorFlow Hub," *tftHub.dev*. <https://tftHub.dev/tensorflow/efficientdet/lite0/detection/1>.